



ASSIGNMENT - 1

SUBJECT: COMPUTER NETWORKS

CODE: 24CST-339

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Q1. Static Routing Topology Design

Objective: Develop a Static Routing topology using 4 Routers, 4 Switches, and 8 PCs; assign IP addresses to all devices, configure static routes between networks, and clearly label all LAN/WAN Network IDs and IP addresses in text format.

Network Topology (Cisco Packet Tracer)

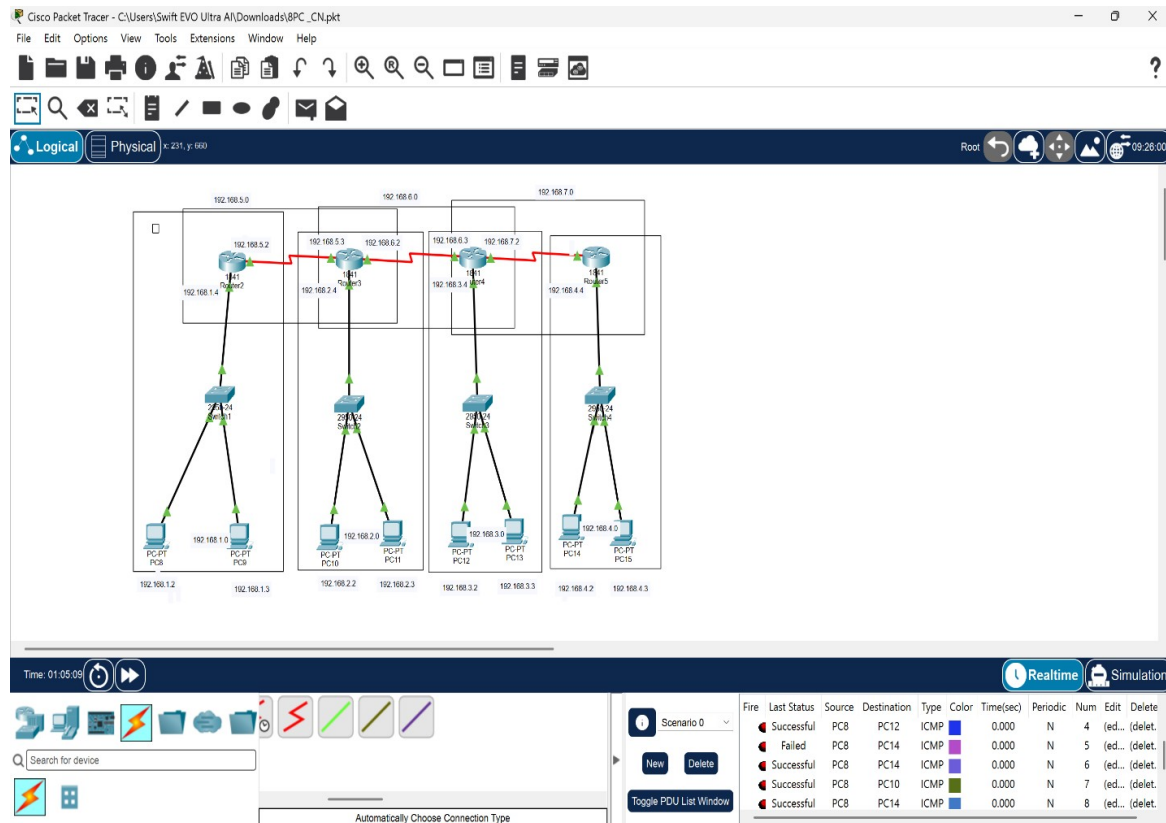


Fig 1.1 — Static routing topology in Cisco Packet Tracer (4 Routers, 4 Switches, 8 PCs)

Network IDs

- LAN 1 Network ID: 192.168.1.0/24
- LAN 2 Network ID: 192.168.2.0/24
- LAN 3 Network ID: 192.168.3.0/24
- LAN 4 Network ID: 192.168.4.0/24
- Router2 ↔ Router3 Network ID: 192.168.5.0/24
- Router3 ↔ Router4 Network ID: 192.168.6.0/24
- Router4 ↔ Router5 Network ID: 192.168.7.0/24

PC IP Addresses

Device	IP Address
PC8	192.168.1.2
PC9	192.168.1.3
PC10	192.168.2.2
PC11	192.168.2.3
PC12	192.168.3.2

PC13	192.168.3.3
PC14	192.168.4.2
PC15	192.168.4.3

Router Interface IP Addresses

Router	Interface	IP Address
Router2	LAN Interface	192.168.1.4
	Serial Interface to Router3	192.168.5.2
Router3	LAN Interface	192.168.2.4
	Serial Interface to Router2	192.168.5.3
	Serial Interface to Router4	192.168.6.2
Router4	LAN Interface	192.168.3.4
	Serial Interface to Router3	192.168.6.3
	Serial Interface to Router5	192.168.7.2
Router5	LAN Interface	192.168.4.4
	Serial Interface to Router4	192.168.7.3

Default Gateways

LAN	Default Gateway
LAN 1 Gateway	192.168.1.4
LAN 2 Gateway	192.168.2.4
LAN 3 Gateway	192.168.3.4
LAN 4 Gateway	192.168.4.4

The tables above summarise the Network IDs and IP addresses, in text format, for the static routing topology shown in Fig 1.1.

Q2. Hub vs Switch vs Router

Question: Write the difference between Hub, Switch & Router in tabular format. Explain why Router is the most intelligent and Hub is the least intelligent device. (3 Marks)

Comparison Table

Feature	Hub	Switch	Router
Definition	Basic device that connects multiple devices and broadcasts data to all ports.	Connects devices within a LAN and forwards data only to the intended device.	Connects different networks and routes data packets between them.
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Data Forwarding	Broadcasts data to all connected devices	Sends data only to the destination device	Routes data to the correct network
Address Used	No address table	Uses MAC Address	Uses IP Address
Collision Domain	One collision domain for all ports	Separate collision domain for each port	Separate collision domain for each interface
Broadcast Domain	One	One (unless VLANs are used)	Different broadcast domains
Speed	Slow	Faster than Hub	Fastest and most efficient
Security	Very Low	Moderate	High
Intelligence Level	Least Intelligent	Moderately Intelligent	Most Intelligent
Example Use	Small or old networks	LAN networks in offices and colleges	Internet and inter-network communication

Why Router is the Most Intelligent Device

A router is considered the most intelligent networking device because:

1. It uses IP addresses to identify source and destination networks.
2. It makes routing decisions and selects the best path for data transmission.
3. It connects different networks, such as LANs, WANs, and the Internet.
4. It maintains routing tables to determine the optimal route.
5. It reduces unnecessary traffic by forwarding packets only to the required network.
6. It provides advanced features such as NAT, DHCP, Access Control Lists (ACLs), firewall functions, and dynamic routing protocols.

Example: When a computer in 192.168.1.0/24 wants to communicate with a computer in 192.168.4.0/24, the router analyses the destination IP address and forwards the packet through the correct route.

Why Hub is the Least Intelligent Device

A hub is considered the least intelligent networking device because:

1. It does not maintain any address table.
2. It cannot identify the destination device.
3. It broadcasts every received packet to all ports.
4. It creates unnecessary network traffic.
5. It causes more collisions and reduces network efficiency.
6. It has no security or filtering capability.

Example: If PC1 sends data to PC2, the hub sends the same data to all connected devices, even though only PC2 needs it. All other devices receive and then discard the packet.

Conclusion

A Hub simply broadcasts data without making any decisions, making it the least intelligent device. A Switch improves network efficiency by using MAC addresses to forward data only to the intended device. A Router is the most intelligent device because it uses IP addresses, routing tables, and advanced networking features to connect and manage communication between different networks efficiently and securely.

Q3. Random Access Protocols

Question: What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also explain the difference between CSMA/CD and CSMA/CA. (3 Marks)

What are Random Access Protocols?

Random Access Protocols are Medium Access Control (MAC) protocols that allow multiple devices to share the same communication channel. In these protocols, any device can transmit data whenever it has data to send, without waiting for a fixed schedule.

Since multiple devices may attempt to transmit at the same time, collisions can occur. Random access protocols provide mechanisms to detect, avoid, or recover from collisions.

Need for Random Access Protocols

1. Allow multiple devices to share a common communication medium.
2. Improve utilization of network resources.
3. Reduce communication delays.
4. Provide fair access to all devices.
5. Handle collisions efficiently in shared networks.

1. ALOHA

ALOHA is the simplest random access protocol. A station transmits data whenever it is ready, without checking whether the channel is free.

Diagram:

```
Station A -----> Data
Station B -----> Data
(Collision occurs if both transmit together)
```

Example:

If Computer A and Computer B send data at the same time, their packets collide. Both computers wait for a random time and then retransmit.

Advantages

- Simple implementation
- No channel sensing required

Disadvantages

- High collision rate
- Low efficiency

2. Slotted ALOHA

Slotted ALOHA improves ALOHA by dividing time into slots. A device can transmit only at the beginning of a time slot.

Diagram:

```
| Slot 1 | Slot 2 | Slot 3 | Slot 4 |
      ^ Data transmission starts here
```

Example:

If a station has data to send, it waits until the next slot begins and then transmits.

Advantages

- Fewer collisions than pure ALOHA
- Better efficiency

Disadvantages

- Requires synchronization of all stations

3. CSMA (Carrier Sense Multiple Access)

Before transmitting, a device listens to the channel. If the channel is busy, it waits. If the channel is free, it transmits.

Diagram:

```
Listen Channel --> Is Channel Free?  
Yes --> Transmit  
No --> Wait
```

Example:

A computer checks whether another device is already sending data. If the channel is free, it starts transmission.

Advantages

- Reduces collisions
- Better channel utilization

Disadvantages

- Collisions can still occur due to propagation delay

4. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

Used mainly in traditional Ethernet networks. A device listens before transmitting and continues monitoring the channel while transmitting. If a collision occurs, transmission stops immediately.

Diagram:

```
Listen --> Transmit --> Detect Collision?  
Yes --> Stop --> Wait Random Time --> Retransmit
```

Example:

Two computers begin transmitting simultaneously. Both detect the collision, stop transmission, wait for a random period, and then retransmit.

5. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Used mainly in wireless networks (Wi-Fi). Since wireless devices cannot easily detect collisions, they try to avoid them by waiting and sending control messages before transmitting.

Diagram:

```
Listen Channel --> Channel Free? --> Yes  
--> Send RTS --> Receive CTS --> Transmit Data  
(RTS = Request To Send, CTS = Clear To Send)
```

Example:

Before sending data, a Wi-Fi device requests permission from the access point. After receiving approval, it transmits.

Difference Between CSMA/CD and CSMA/CA

Feature	CSMA/CD	CSMA/CA
Full Form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Working Principle	Detects collisions after they occur	Tries to prevent collisions before they occur
Used In	Wired Ethernet Networks	Wireless Wi-Fi Networks
Collision Handling	Detect and retransmit	Avoid and then transmit
Mechanism	Collision Detection	RTS/CTS and Backoff Mechanism
Efficiency	High in wired networks	Better for wireless environments
Transmission	Device transmits and monitors channel	Device waits before transmitting
Complexity	Simple	More complex